THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which lens gives a virtual, erect and magnified image? 1

(a) Convex with object at 2F **(b)** Concave with object at F
(c) Convex with object between F and O **(d)** Concave with any object

2. For a spherical mirror of focal length f, magnification m = -2 means 1

(a) real inverted image twice as big **(b)** virtual erect image twice as big
(c) real inverted image half as big **(d)** virtual inverted image

3. If the refractive index of A w.r.t. B is 1.5, the refractive index of B w.r.t. A is 1

(a) 2/3 **(b)** 3/2 **(c)** 1.5 **(d)** 3

4. A lens forms an inverted image of the same size as the object at 20 cm from the lens. Its focal length is 1

(a) 40 cm **(b)** 10 cm **(c)** 20 cm **(d)** 5 cm

5. If the radius of curvature of a concave mirror is 60 cm, an object at 30 cm gives an image at 1

(a) 15 cm **(b)** 30 cm **(c)** 60 cm **(d)** infinity

6. Where does a fish appear to be when viewed from above? 1

(a) At the same depth
(b) Nearer to the surface than the actual position
(c) Deeper than the actual position
(d) On the surface

7. A convex lens has f = 25 cm. Its power is 1

(a) +25 D **(b)** -4 D **(c)** +0.04 D **(d)** +4 D

8. Which of these is a real image? 1

(a) Image in a plane mirror **(b)** Image on a screen from a concave mirror
(c) Image in a convex mirror **(d)** Image in a magnifying glass

9. For a ray emerging from a rectangular slab, which statement is true? 1

(a) Angle of emergence = 0 **(b)** Angle of emergence < angle of incidence
(c) Angle of emergence > angle of incidence **(d)** Angle of emergence = angle of incidence

10. The nature of image formed by a concave lens, when the object is at infinity, is 1

(a) at F, virtual, point-sized **(b)** at F, real
(c) at infinity **(d)** at 2F

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. An object 2 cm tall is at 10 cm from a convex lens of f = 5 cm. Find the image size. 2

12. Why does a ray bend towards the normal in a denser medium? 2

13. An object is at 15 cm from a convex lens of f = 10 cm. Find the image distance. 2

14. Why is a coin in a beaker of water apparently raised? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The refractive indices of water and glass are 1.33 and 1.5. Find the refractive index of glass with respect to water. 3

16. Explain why an object between F and O of a convex lens gives an erect, magnified, virtual image, with the help of the lens formula. 3

17. A ray of light parallel to the principal axis strikes a convex mirror. Describe the reflected ray. 3

OR

A convex lens of f = 20 cm forms a real image at 60 cm. Find the object distance and magnification. 3

18. Find the focal length of a concave mirror that forms a virtual image 30 cm behind it of an object 10 cm from it. 3

OR

How do the signs of u, v, f help to know the nature of the image in the mirror formula? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A convex lens of f = 10 cm is used as a magnifying glass. Find the range of object distances. For u = 6 cm, find the image distance and magnification. 5

OR

Explain how a concave mirror forms an image of the same size as the object; calculate the object distance for a mirror of radius of curvature 36 cm. 5

20. Find the focal length, power and nature of a lens that forms a virtual image three times smaller than an object 30 cm away. 5

OR

State the lens formula, the magnification formula and derive the position when u = -2f for a convex lens. Hence state the nature of the image. 5

END OF QUESTION PAPER — Total: 40 Marks

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